UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 7 printed pages

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QUESTION 1

a) Solve the compound inequality -3 < 3x - 1 < 2. Express the answer in interval notation.

(4 marks)



Figure: solution on a number line.

b) Solve the inequality |2 - 5x| > 7. Express the answer in interval notation and on a number line.

(5 marks)

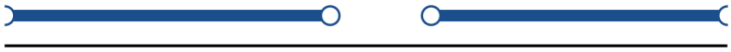


Figure: solution on a number line.

c) Sketch y = -3|x + 4| - 3. State the vertex, the y-intercept and any x-intercepts.

(5 marks)

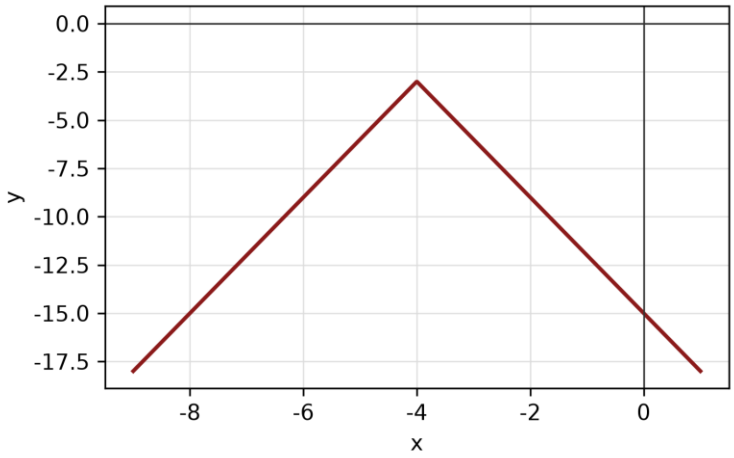


Figure: absolute-value graph.

QUESTION 2

a) Given f(x) = (3 - 5x)/(x - 4) and g(x) = √(3x - 5), state the domain of each function.

(4 marks)

b) Apply graph-transformation techniques to sketch $y = -2(x - 4)^2 + 5$. State the vertex and the y-intercept.

(5 marks)

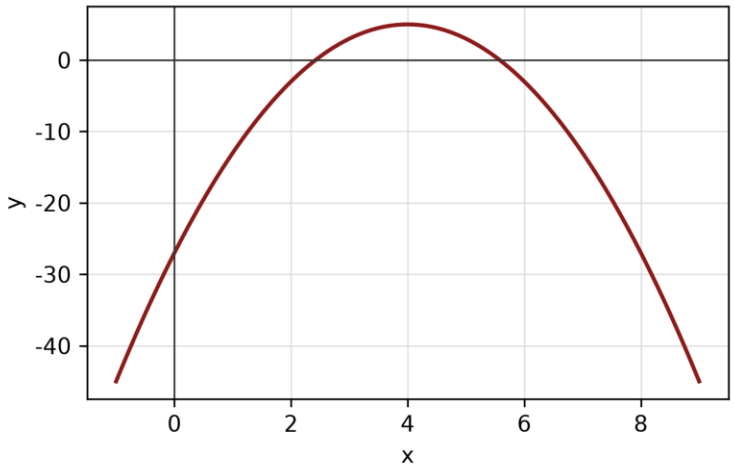


Figure: transformed graph.

c) Write the equation of the circle with centre (4, 1) and radius 2. Hence write its general equation and sketch the circle.

(5 marks)

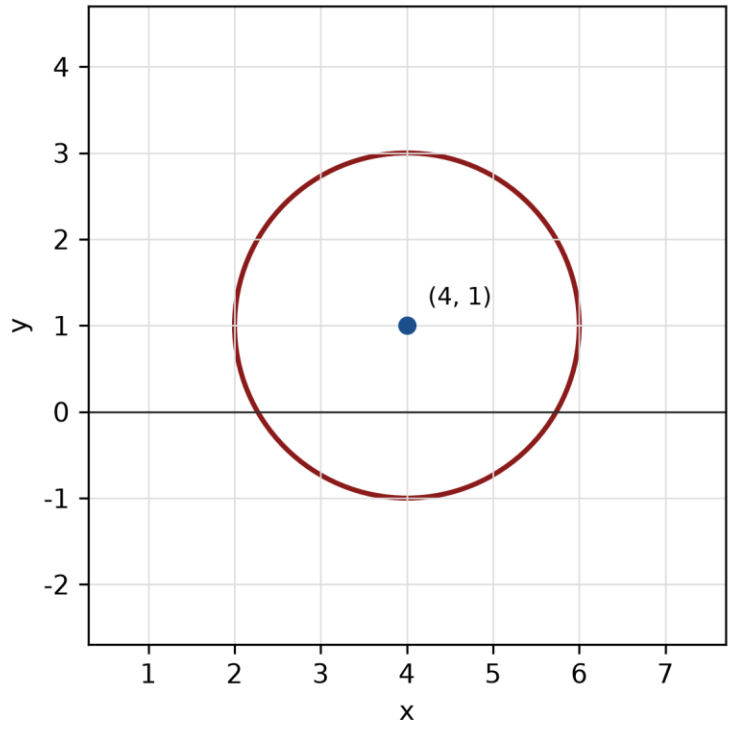


Figure: circle.

QUESTION 3

a) Given $z_1 = 2 + 4i$ and $z_2 = 3 + 2i$, express z_1/z_2 in the form a + bi.

(5 marks)

b) Find the modulus and argument of $z = -5 + 12i$. Give the argument in degrees correct to one decimal place.

(5 marks)

c) Factorise $f(x) = x^3 - 2x^2 - 9x + 18$ completely.

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(5 marks)

QUESTION 4

a) Solve the system of equations simultaneously:
$-3x + 3y + z = -14$
$2x + 2y - 3z = 2$
$2x + 3y + 3z = -13$

(5 marks)

b) Graph the system of inequalities below and shade the feasible region. State the vertices of the region.
$x \geq 0, y \geq 0, 1x + 3y \leq 6.$

(5 marks)

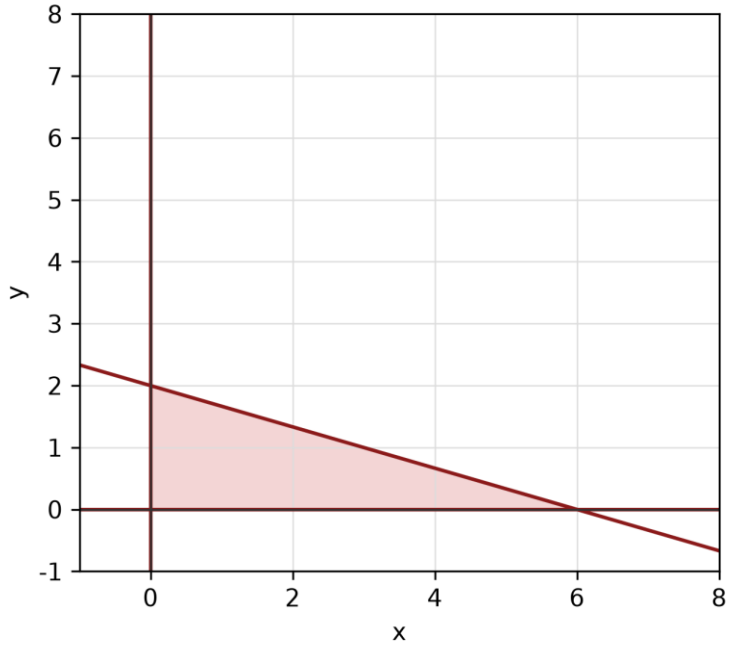


Figure: feasible region.

c) Find the equation of the line with x-intercept 4 and y-intercept -2.

(4 marks)

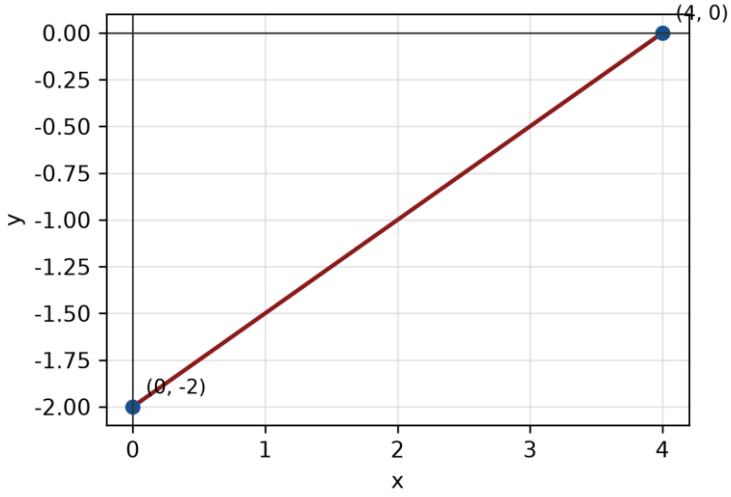


Figure: line and intercepts.

QUESTION 5

a) For y = 2 sin(1x + 30) + 2, state the amplitude, period and phase shift, then sketch the graph for -180° ≤ x ≤ 180°.

(5 marks)

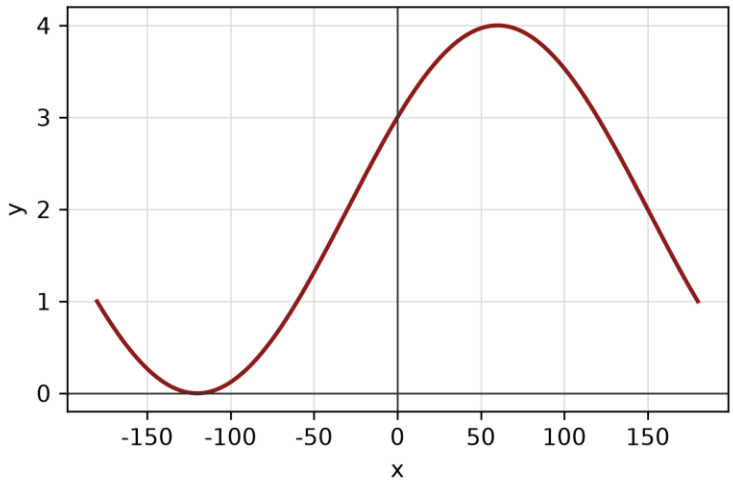


Figure: trigonometric graph.

b) A circle has radius 6 cm. Find the length of the arc subtended by a central angle of 30°.

(4 marks)

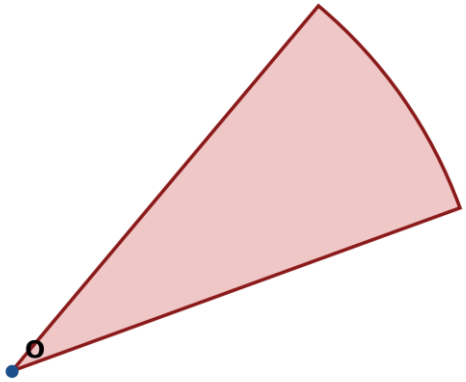


Figure: arc and sector.

c) In triangle ABC, b = 9 cm, c = 12 cm and A = 120°. Use the cosine rule to find a.

(4 marks)

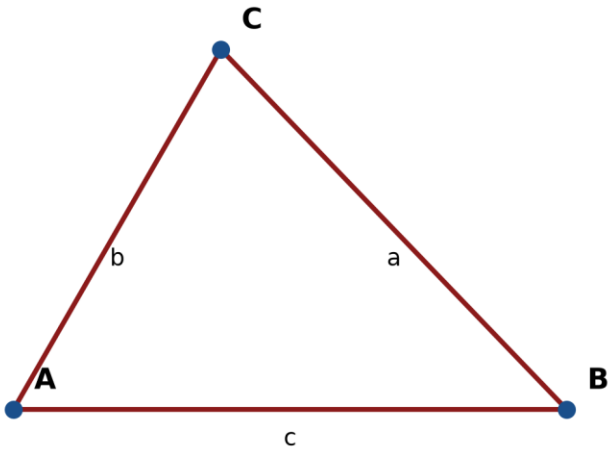


Figure: triangle ABC.

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$

$\cos 2A = \cos^2 A - \sin^2 A$

$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$

$\sin 2A = 2 \sin A \cos A$

$\tan^2 A + 1 = \sec^2 A$

$\tan 2A = 2 \tan A / (1 - \tan^2 A)$

Arc length, $s = r\theta$

Area of sector = $\frac{1}{2}r^2\theta$

$a / \sin A = b / \sin B = c / \sin C$

$a^2 = b^2 + c^2 - 2bc \cos A$

Area of triangle = $\frac{1}{2}ab \sin C$

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